The φ-Clock Universe

VR & AI

Abstract

We give a complete first-principles formalism for a temporal universe driven by a single bi-infinite binary register, a single tick at k=0, and the unique value-preserving local rewrite $11 \to 100$. All derivations use only the golden identity $\varphi^2 = \varphi + 1$ (equivalently $1 + \varphi^{-1} = \varphi$) and this rewrite. Negative indices are read through a purely temporal inversion lens into outer scale-channel labels (R=φ^m, r=$(2m)\,(\mathrm{mod}\ 3)$), yielding conserved fast content $Q^- = \sum_{m\geq 1} U_{-m}\varphi^{-m}$. Detuned unity admits a φ-log spiral envelope; the per-bit one-turn spiral-action is linear in φ^{-m}, so the total action is a constant multiple of Q^-. We define species (fast masks), a minimal read-out process into a uniform potential (instantiation), the expected instantiation rate $\lambda_S = \sum S(m)U_{-m}\varphi^{-2m}$, a branching readout that yields exponential growth ("big bang") without changing the register, and an algebraic binding vs. scatter rule via φ-closure (worked example). A Z_3 spinor-like internal cycle arises from residues $r = (2m)\,(\mathrm{mod}\ 3)$ on depth triplets $\{m_0, m_0{+}2, m_0{+}4\}$, with phase advances gated by rung statistics. We close with a minimal field picture and a simple complexity bound for the tick.

1 Axioms and admissible class

Let $\varphi = \frac{1+\sqrt{5}}{2}$ so $\varphi^2 = \varphi + 1$ and $1 + \varphi^{-1} = \varphi$.

Definition 1 (Register and admissibility)**.** *A register is a bi-infinite binary string*

$$U = (\ldots, U_{-1} \ \mid\ U_0, U_1, U_2, \ldots), \qquad U_k \in \{0, 1\},$$

with the single constraint $U_k U_{k+1} \neq 11$ *(*no adjacent ones*).*

Definition 2 (Tick and normalization)**.** *The unique value-preserving local rewrite is*

$$\underline{11} \ \longrightarrow\ \underline{100} \qquad \textit{at indices } (k{+}1, k) \mapsto (k{+}2, k{+}1, k), \tag{1}$$

which restates $\varphi^{k+1}{+}\varphi^k = \varphi^{k+2}$*. A tick injects* 1 *at* $k = 0$ *and normalizes:*

$$U \ \mapsto\ \mathsf{Norm}\big(U + \mathbf{e}_0\big), \qquad (\mathbf{e}_0)_0 = 1,\ (\mathbf{e}_0)_k = 0\ (k \neq 0). \tag{2}$$

Remark 1 (Why $11 \to 100$ and why it halts)*.* Let $E_q(U) := \sum_k q^k U_k$ for any $q \in (1, \varphi)$. Then under (1) we have

$$E_q(100) - E_q(11) = q^{k+2} - (q^{k+1}{+}q^k) = q^k(q^2 - q - 1) \ <\ 0,$$

so repeated rewrites strictly decrease E_q, hence normalization halts.

We work in a natural class that ensures well-defined functionals on the fast tail.

Definition 3 (Admissible class)**.** $\mathcal{C} := \big\{U \textit{ admissible} \ :\ \#\{k \geq 0 : U_k = 1\} < \infty,\ \sum_{m\geq 1} U_{-m}\varphi^{-m} < \infty\big\}$.

2 Uniqueness and locality

Normalization order (determinism). To resolve the 111 critical overlap at indices $(-1, 0, 1)$ we *fix* normalization to apply $11 \to 100$ from *higher to lower* indices (rightmost-first scan). This makes the post-tick normal form independent of rule order. With this choice, a boundary rewrite across $(-1, 0)$ can occur at the first tick only if $U_{-1} = 1$ and $U_1 = 0$ just before the tick.

Theorem 1 (Normal form exists and is unique)**.** *Each tick (insertion at $k = 0$) followed by rightmost-first normalization halts and yields a unique normal form.*

Sketch. Halting: the potential $E_q(U) = \sum_k U_k q^k$ with $1 < q < \varphi$ strictly decreases on $11 \to 100$. Local confluence: the only critical pair is 111, which our rightmost-first order resolves deterministically by firing $(0, 1)$ before $(-1, 0)$. By Newman's lemma (termination + local confluence), normalization is confluent; thus the normal form is unique. □

Proposition 1 (Tick locality)**.** *Under a tick at $k = 0$ and subsequent normalization, digits with $k \leq -2$ never change.*

Proof. Each rewrite replaces $(k, k{+}1)$ by $(k{+}2, k{+}1, k)$, so the minimum index touched is non-decreasing. Starting at $k = 0$, no chain of carries can cross below -1. □

Tickability. We restrict ticks to states with $U_0 = 0$ (the slow prefix begins with a 0); this avoids intermediate digits "2" during insertion.

Lemma 1 (Uniqueness of normal form (Zeckendorf-like))**.** *If U, V are admissible and $\sum_k U_k \varphi^k = \sum_k V_k \varphi^k$, then $U = V$.*

Remark 2 (Boundary probability: ensemble vs. time)*.* A boundary event at the first tick occurs iff $U_{-1} = 1$ (and, with the rightmost-first rule, $U_1 = 0$). Under the Parry (max-entropy) measure on the golden-mean subshift, the marginal $\Pr[U_{-1} = 1] = \pi_1 = \frac{1}{1+\varphi^2}$. Along any single trajectory the boundary can occur at most once, so its time-average frequency tends to zero; π_1 is an *ensemble* probability over initial states. Under a stationary ergodic measure, the boundary event is a zero-frequency tail event along a single trajectory; probabilities are ensemble quantities.

3 Projection lens and conserved fast amount

We do *not* introduce space. The lens is a temporal relabeling of negative bits. For each $m \geq 1$, if $U_{-m} = 1$ we define an outer feature

$$(R_m, r_m) := \big(\varphi^m,\ (2m) \bmod 3\big). \tag{3}$$

Define the fast amount

$$\boxed{Q^-(U) := \sum_{m \geq 1} U_{-m}\, \varphi^{-m}} = \sum_{(R_m, r_m)} \frac{1}{R_m}. \tag{4}$$

Lemma 2 (Boundary and conservation)**.** *Any rewrite $11 \to 100$ with $k \leq -2$ leaves Q^- unchanged. A rewrite at $k = -1$ reduces Q^- by exactly φ^{-1}.*

Remark 3 (Rarity of boundary events)*.* In a single trajectory the boundary event can happen at most once (it requires $U_{-1} = 1$ just before the tick; afterwards $U_{-1} \leftarrow 0$). Under the Parry (max-entropy) measure on the golden-mean subshift, $\Pr[U_{-1} = 1] = \frac{1}{1+\varphi^2}$; along a single trajectory its time-average frequency tends to zero.

4 Detuned unity as a φ-log temporal spiral

Let $r(\theta) = r_\star e^{b\theta}$ be a logarithmic spiral. Its arc length between θ_0 and θ_1 is

$$L = \int_{\theta_0}^{\theta_1} \sqrt{r^2 + \left(\frac{dr}{d\theta}\right)^2}\, d\theta = \sqrt{1+b^2} \int r(\theta)\, d\theta = \frac{\sqrt{1+b^2}}{b}\, r(\theta_0)\big(e^{b(\theta_1-\theta_0)} - 1\big). \tag{5}$$

Choose growth per quarter turn $g = e^{b\pi/2}$ so over one full turn $e^{b(\theta_1-\theta_0)} = g^4$. Associate a fast bit at $-m$ with starting diameter $D_m = \varphi^{-m}$ (so $r(\theta_0) = \frac{1}{2}\varphi^{-m}$) and orientation $\sigma \in \{+1, -1\}$ with $g = \varphi^\sigma$. Then the *per-bit* one-turn spiral-action is

$$L_m^{(\sigma)} = \frac{1}{2}\, \frac{\sqrt{1+b_\sigma^2}}{b_\sigma}\, \big(\varphi^{4\sigma} - 1\big)\, \varphi^{-m} =: C_\sigma\, \varphi^{-m}, \qquad b_\sigma = \tfrac{2}{\pi} \ln(\varphi^\sigma). \tag{6}$$

Summing over the fast tail,

$$\boxed{L_{\text{turn}}^{(\sigma)}(U) = \sum_{m\geq 1} U_{-m}\, L_m^{(\sigma)} = C_\sigma\, Q^-(U)}, \tag{7}$$

a constant multiple of Q^- (changes only at the single boundary by $\Delta L^{(\sigma)} = -C_\sigma \varphi^{-1}$).

5 Slow rungs and scaling

A tick reaches slow rung n iff the pre-tick slow prefix equals the alternating word $(01)^n$. Under a natural stationary measure on admissible slow prefixes (the golden-mean subshift), the probability of $(01)^n$ is $\propto \varphi^{-2n}$, yielding

$$\boxed{f_n \;\propto\; \varphi^{-2n} f_0, \qquad T_n \;\propto\; \varphi^{2n} T_0}. \tag{8}$$

Remark 4 (Measure remark). The proportionality is asymptotic under any stationary, ergodic measure compatible with no-adjacency; for a fixed initial U it gives a robust envelope for long-run frequencies.

Markov construction for rung statistics

Let G be the golden-mean subshift forbidding 11. The 2×2 adjacency is $A = \left(\begin{smallmatrix} 1 & 1 \\ 1 & 0 \end{smallmatrix}\right)$ with Perron root $\lambda = \varphi$ and right/left eigenvectors $v = (\varphi, 1)$, $w = (1, \varphi^{-1})$.

The Parry transition matrix is $P = \left(\begin{smallmatrix} \varphi^{-1} & \varphi^{-2} \\ 1 & 0 \end{smallmatrix}\right)$. Its stationary distribution is

$$\pi = \left(\frac{\varphi^2}{1+\varphi^2}, \frac{1}{1+\varphi^2}\right),$$

so for the alternating prefix $(01)^n$ we have

$$\Pr[(01)^n] = \pi_0\, (P_{01}P_{10})^n = \frac{\varphi^2}{1+\varphi^2}\, \varphi^{-2n} \propto \varphi^{-2n}.$$

Hence the expected waiting time to reach rung n scales as φ^{2n}.

6 Species schema (genotype)

A *species* is a fast-depth/channel mask: choose $\mathcal{R}_S \subseteq \{0,1,2\}$ and $\mathcal{M}_S \subseteq \mathbb{N}_{\geq 1}$; set $S(m) = \mathbf{1}\{\, m \in \mathcal{M}_S \;\&\; (2m) \bmod 3 \in \mathcal{R}_S \,\}$. Define observables

$$\text{Multiplicity:} \quad N_S(U) = \sum_{m\geq 1} S(m)\, U_{-m}, \tag{9}$$

$$\text{Weight:} \quad W_S(U) = \sum_{m\geq 1} S(m)\, U_{-m}\, \varphi^{-m}, \tag{10}$$

$$\text{Channels:} \quad \mathbf{C}_S(U) = (C_0, C_1, C_2),\ C_j = \sum_{m\geq 1} S(m)\, U_{-m}\, \mathbf{1}\{(2m) \bmod 3 = j\}, \tag{11}$$

$$\text{Central depth/time-scale:} \quad \bar{m}_S = \frac{\sum m\, S(m)\, U_{-m}\, \varphi^{-m}}{\sum S(m)\, U_{-m}\, \varphi^{-m}}, \quad T_S \propto \varphi^{2\bar{m}_S} T_0, \tag{12}$$

$$\text{Spiral-action:} \quad L_S^{(\sigma)} = C_\sigma\, W_S(U). \tag{13}$$

By Proposition **??**, these are constants of motion except for the one-off boundary at m=1 (if present in S).

7 Instantiation in a uniform potential (phenotype)

The outside *foam* is a uniform set of temporal slots $\mathcal{F} = \{(R > 1, r \in \{0,1,2\})\}$. A tick at time τ reaches a random rung n_τ; it *instantiates* all fast bits at depths $m \leq n_\tau$:

$$\mathcal{I}_\tau = \{\, (R_m = \varphi^m,\ r_m) \ :\ U_{-m} = 1,\ m \leq n_\tau \,\}. \tag{14}$$

Under the stationary measure, $\Pr[n_\tau \geq m] = \varphi^{-2m}$, hence the expected per-tick instantiation count for species S is

$$\boxed{\lambda_S = \sum_{m\geq 1} S(m)\, U_{-m}\, \varphi^{-2m}}, \qquad \mathbb{E}\, N_S(T) = T\, \lambda_S. \tag{15}$$

Branching readout (exponential phase, no change to U). If each instantiation becomes a new reader with probability α and each reader produces on average $\lambda_* = \sum_{m\geq 1} U_{-m}\varphi^{-2m}$ instantiations per tick, then the expected reader count obeys

$$\mathbb{E}\, \mathcal{R}_{t+1} = (1 + \alpha\, \lambda_*)\, \mathbb{E}\, \mathcal{R}_t \quad \Rightarrow \quad \mathbb{E}\, \mathcal{R}_t = \mathcal{R}_0 (1 + \alpha\, \lambda_*)^t, \tag{16}$$

i.e. an exponential "big bang" phase arising purely from readout replication.

8 Co-instantiation and binding via φ-closure

Greedy φ-closure and uniqueness. Given a finite multiset of depths $M \subset \mathbb{N}$ with total weight $W = \sum_{m\in M} \varphi^{-m}$, let $m_{\text{eff}} = \max\{m : \varphi^{-m} \leq W\}$, $N = \lfloor W\varphi^{m_{\text{eff}}} \rfloor$, and $\gamma = W - N\varphi^{-m_{\text{eff}}} \in [0, \varphi^{-m_{\text{eff}}})$. Then

$$W = N\varphi^{-m_{\text{eff}}} + \gamma,$$

is unique: shifting m_{eff} up would force $\gamma < 0$; down would make $\gamma \geq \varphi^{-m_{\text{eff}}}$. We bind N composites at depth m_{eff} and route γ to a massless cascade.

The expected *co-instantiation* rate of S_1 and S_2 is

$$\boxed{\Lambda_{12} = \sum_{m\geq 1} S_1(m)\, S_2(m)\, U_{-m}\, \varphi^{-2m}}. \tag{17}$$

Upon co-instantiation at the same depth m (with channels fixed by $r = (2m) \bmod 3$), we adopt a minimal algebraic outcome:

Interaction Rule (IR). Given two co-instantiated depths, perform the greedy φ-closure on their total weight. If the resulting multiplicity $N \geq 1$, declare binding with multiplicity N at depth m_{eff} and record the remainder γ in the massless channel. If $N = 0$, declare scatter.

Worked example. Depths m and $m{+}2$ (non-adjacent in U) have weights φ^{-m} and $\varphi^{-(m+2)}$:

$$\varphi^{-m} + \varphi^{-(m+2)} = (\varphi + \varphi^{-1})\,\varphi^{-(m+1)} = \sqrt{5}\,\varphi^{-(m+1)}. \tag{18}$$

We declare a *composite* at depth $m{+}1$ (channel rotates by $+2 \bmod 3$) plus a radiative remainder of $(\sqrt{5}-1)\varphi^{-(m+1)}$ in γ:

$$A_m + B_{m+2} \;\longrightarrow\; C_{m+1} + \gamma.$$

Action bookkeeping: $L^{(+)}(A_m) + L^{(+)}(B_{m+2}) = L^{(+)}(C_{m+1}) + L^{(+)}(\gamma)$ by linearity in (6).

9 Z_3 spinor-like cycle (explicit construction)

Let a spinor species use depths $\mathcal{M}_{\text{sp}} = \{m_0, m_0{+}2, m_0{+}4\}$ (respecting no-adjacency). Channels cycle: $r(m_0{+}2k) = (2m_0{+}4k) \bmod 3 = r(m_0) + k \pmod 3$. Define an internal phase $p \in \{0,1,2\}$ as the index of the most recently activated component; when a tick reaches the next component (depth increases by 2), the phase advances

$$p \;\mapsto\; p+1 \pmod 3. \tag{19}$$

The probability per tick to activate depth $m_0{+}2k$ is $\varphi^{-2(m_0+2k)}$, hence the expected waiting time for each phase step is

$$\mathbb{E}[\Delta t_k] = \varphi^{2(m_0+2k)}. \tag{20}$$

This gives a Z_3 internal cycle with geometrically increasing periods—spinor-like and purely temporal (not SU(2)).

10 Minimal field picture (optional bridge)

Let $\psi_r(m;t)$ be the accumulated instantiations in the foam. The tick defines a source

$$J_r(m;t) = U_{-m}\,\mathbf{1}\{n_t \geq m\}\,\mathbf{1}\{r = (2m) \bmod 3\}, \tag{21}$$

and a free update $\psi_r(m;t{+}1) = \psi_r(m;t) + J_r(m;t)$. A minimal interaction (matching (18)) is a φ-closure vertex

$$\psi_{r+2}(m{+}1;t{+}1) \;\leftarrow\; \kappa\,\psi_r(m;t)\,\psi_{r+1}(m{+}2;t), \tag{22}$$

with a massless channel γ recording the radiative remainder. No spatial geometry is assumed; this is a discrete, scale-indexed skeleton compatible with field-theoretic language.

Operational distance and synchronization

Define the shared-prefix length $\ell(U,V) = \max\{L:\; U_{-m} = V_{-m}\ \forall m \leq L\}$ and the ultrametric

$$d_\varphi(U,V) = \varphi^{-\ell(U,V)}.$$

An operational estimator from ticks is obtained by the shortest time T^* needed for both registers to exhibit the same rung n; by $\mathbb{E}[T_n] \propto \varphi^{2n}$ we set $\widehat{d}_\varphi \sim (T^*/T_0)^{-1/2}$.

11 Algorithmic complexity

Deterministic tick harness (for simulation)

1. Require $U_0 = 0$; set $U_0 \leftarrow 1$ (tick).
2. For k descending from the largest index with $U_k = 1$ down to -1, apply $11 \to 100$ at $(k-1, k)$ whenever present.
3. Record: whether $(-1, 0)$ fired (boundary), the rung n (length of initial alternating 01), and any species readouts.

This runs in $O(n)$ when the slow prefix is $(01)^n$.

Let the pre-tick slow prefix length be n (maximal $(01)^n$). Normalization performs $O(n)$ rewrites and touches indices $\{0, 1, \ldots, O(n)\}$. Negative indices $k \leq -2$ are never touched; the boundary -1 can be cleared once. Hence a tick has worst-case time $O(n)$ and space $O(n)$ on the slow side.

12 One-page recipe

- **Engine:** single tick at $k = 0$, single rewrite $11 \to 100$.
- **Lens:** fast bit $-m \mapsto (R = \varphi^m,\ r = (2m) \bmod 3)$.
- **Conserved:** $Q^- = \sum U_{-m}\varphi^{-m}$ (changes only by $-\varphi^{-1}$ at $(-1, 0)$).
- **Spiral-action:** $L^{(\sigma)} = C_\sigma Q^-$ with C_σ from (6).
- **Rungs:** $f_n \propto \varphi^{-2n}$, $T_n \propto \varphi^{2n}$ (natural measures).
- **Species:** $(\mathcal{R}_S, \mathcal{M}_S) \Rightarrow N_S, W_S, \mathbf{C}_S, \bar{m}_S, L_S$ (constants of motion).
- **Instantiation:** per-tick rate $\lambda_S = \sum S(m) U_{-m}\varphi^{-2m}$; with branching readers (16) gives exponential growth.
- **Interactions:** co-rate $\Lambda_{12} = \sum S_1 S_2\, U_{-m}\varphi^{-2m}$; bind iff φ-closure to a single depth exists; else scatter.
- **Spinor:** depths $m_0, m_0{+}2, m_0{+}4$ give a Z_3 phase advancing when rung reaches the next depth.

Figure

Consistency and minimalism

Every equation is a consequence of $\varphi^2 = \varphi + 1$ and $11 \to 100$, plus standard calculus for the log-spiral length (5) (choice of units enters only through starting diameter). No extra forces, spaces, or Hamiltonians are assumed; if a spatial metric is ever desired, it must be constructed operationally from synchronization times, not postulated.

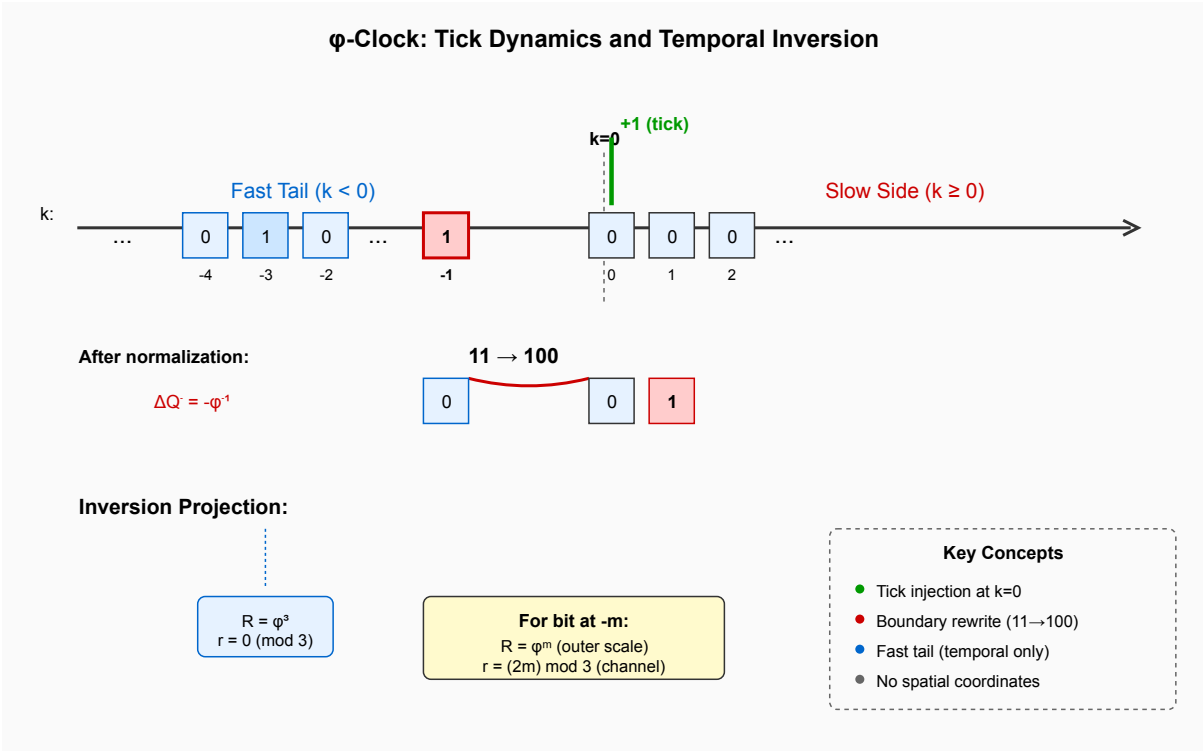

Figure 1: Tick at $k = 0$ and the unique normalization $11 \to 100$ across $(-1, 0)$ remove the nearest fast bit and promote one unit to the slow side, changing Q^- by $-\varphi^{-1}$. The inversion lens re-labels a fast bit at $-m$ as $(R = \varphi^m,\ r = (2m) \bmod 3)$. The view is purely temporal (no spatial postulate).

Appendix: Concrete Examples

A1. A small register and its invariants

Consider the admissible register with fast bits at depths $m = 1$ and $m = 3$ (i.e. $U_{-1} = U_{-3} = 1$) and no slow bits. Then

$$Q^-(U) = \varphi^{-1} + \varphi^{-3} \;=\; 0.618034 + 0.236068 \;=\; 0.854102,$$

$$L^{(+)}_{\text{turn}}(U) = C_+\, Q^- \quad \text{with } C_+ = \tfrac{1}{2}\frac{\sqrt{1+b_+^2}}{b_+}(\varphi^4 - 1) \approx 9.992927 \;\Rightarrow\; L^{(+)}_{\text{turn}} \approx 8.534978,$$

$$L^{(-)}_{\text{turn}}(U) = C_-\, Q^- \quad \text{with } C_- = \tfrac{1}{2}\frac{\sqrt{1+b_-^2}}{b_-}(\varphi^{-4} - 1) \approx 1.457948 \;\Rightarrow\; L^{(-)}_{\text{turn}} \approx 1.245237.$$

(Here $b_\pm = \frac{2}{\pi}\ln(\varphi^\pm)$.) These numbers match Eq. (7): action is linear in $Q^-(U)$.

A2. One tick and the slow rung

Let the pre-tick slow prefix be $(01)^2$, i.e. $(U_0, U_1, U_2, U_3) = (0, 1, 0, 1)$. Injecting a tick at $k = 0$ and applying only $11 \to 100$:

$$\cdots \mid \underline{0,1},0,1 \;+\; \mathbf{e}_0 \;\Rightarrow\; \cdots \mid \underline{1,1},0,1 \;\xRightarrow{11\to100}\; \cdots \mid 0,0,\underline{1,1}$$
$$\xRightarrow{11\to100}\; \cdots \mid 0,0,0,0,1,$$

so the tick reaches *rung* $n = 2$ and leaves a single 1 at index $2n = 4$ (all intermediate slow indices cleared).

A3. Instantiation rate for a species

Let the species mask S select depths $\{1, 3\}$. Then the expected instantiations per tick are

$$\lambda_S = \sum_{m\in\{1,3\}} \varphi^{-2m} = \varphi^{-2} + \varphi^{-6} = 0.381966 + 0.055728 = 0.437694.$$

Over T ticks the expected count is $T\,\lambda_S$. The *rest content* of this species is

$$W_S = \sum_{m\in\{1,3\}} \varphi^{-m} = \varphi^{-1} + \varphi^{-3} = 0.854102, \qquad L_S^{(\sigma)} = C_\sigma\, W_S.$$

A4. Binding vs. scatter by φ-closure (worked)

Two co-instantiated depths m and $m{+}2$ have weights φ^{-m} and $\varphi^{-(m+2)}$. Their sum is

$$\varphi^{-m} + \varphi^{-(m+2)} = (\varphi + \varphi^{-1})\varphi^{-(m+1)} = \sqrt{5}\,\varphi^{-(m+1)}.$$

Thus we declare a composite at depth $m{+}1$ plus a *massless* remainder γ of size $(\sqrt{5}-1)\varphi^{-(m+1)}$. For $m = 2$ one gets numerically

$$\begin{aligned} \varphi^{-m} + \varphi^{-(m+2)} &= 0.527864, \\ \text{composite weight at } m{+}1 &= \varphi^{-(m+1)} = 0.236068, \\ \gamma \text{ (radiative remainder)} &= (\sqrt{5}-1)\varphi^{-(m+1)} = 0.291796. \end{aligned}$$

Action bookkeeping follows from linearity: $L^{(+)}$ of inputs equals $L^{(+)}$ of outputs.

A5. $\mathbf{Z}_3$ spinor cycle (explicit)

Choose $m_0 = 2$, so the triplet depths are $\{2, 4, 6\}$ with residues $r = \{1, 2, 0\} = \{(2m) \bmod 3\}$. Each activation advances the internal phase $p \mapsto p + 1 \pmod 3$. Expected waiting times per phase step are

$$\mathbb{E}[\Delta t_k] = \varphi^{2(m_0+2k)} = \{6.854102,\ 46.978714,\ 321.996894\}.$$

A6. Branching readout ("big bang" phase)

If every instantiation becomes a new reader with probability $\alpha = 0.120000$ and a reader's mean per-tick output is $\lambda_* = \sum_{m=1}^{4} \varphi^{-2m} = 0.604878$, then the expected reader count grows as

$$\mathbb{E}\,\mathcal{R}_t = \mathcal{R}_0\,(1 + \alpha\lambda_*)^t = \mathcal{R}_0\,(1.072585)^t.$$

This is exponential, while the register and its law remain unchanged.